# DATA MINING

# CS F415




# Customer Segmentation using Clustering: An Approach using Autoencoders


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# Problem Statement

## Abstract:

Customer segmentation is an essential step for companies looking to personalize marketing and improve customer relationship management. In this project, we introduce an innovative method that involves using autoencoders to perform non-linear dimensionality reduction before clustering. We use K-Means, Gaussian Mixture Models (GMM), and DBSCAN on the reduced representations to reveal insightful customer segments. This approach utilizes the UK Online Retail Dataset and aims to enhance cluster quality and interpretability by identifying intricate patterns in customer behavior.

## Dataset:

- UK Online Retail Dataset (from UCI Machine Learning Repository): 541,909 customer transactions with features used in RFM (Recency, Frequency, Monetary) analysis.
- Dataset Features:
    - InvoiceNo: Unique ID for each transaction. Transactions with a 'C' prefix indicate cancellations.
    - StockCode: Unique product code.
    - Description: Name of the product.
    - Quantity: Number of units per transaction.
    - InvoiceDate: Timestamp of the transaction.
    - UnitPrice: Price per unit in sterling.
    - CustomerID: Unique identifier for each customer.
    - Country: Country where the customer resides.

## Implementation Roadmap:

### 1. Data Preprocessing:

- Feature Engineering: Computed RFM scores to summarize customer purchasing behavior.
- Data Cleaning: Removed incomplete records and handled transaction cancellations.
- Feature Scaling: Standardized RFM features to ensure balanced input to the autoencoder and clustering algorithms.

### 2. Non-Linear Dimensionality Reduction with Autoencoders

Classic linear methods such as PCA are unable to embed intricate, non-linear patterns in customer data. To overcome this, autoencoders—based on deep learning—were utilized to learn compressed, informative representation even under the presence of high dimensions and noise. The model was trained with mean squared error loss for reconstruction, and early stopping was used to avoid overfitting. The resulting encoded representations were next utilized as input for the next clustering algorithms.

**Encoder (non-linear transformation):**

$$\mathbf{z} = f_\phi(\mathbf{x}) = \mathrm{ReLU}(\mathbf{W}_e\mathbf{x} + \mathbf{b}_e)$$

**Decoder (Reconstruction):**

$$\hat{\mathbf{x}} = g_\theta(\mathbf{z}) = \sigma(\mathbf{W}_d\mathbf{z} + \mathbf{b}_d)$$

where,

$\mathbf{x} \in \mathbb{R}^d$ : Original input vector (RFM features + transaction metadata)

$\mathbf{h} \in \mathbb{R}^k$ : Latent representation ($k \ll d$)

$\mathbf{W}_e$, $\mathbf{W}_d$ : Weight matrices

$\mathbf{b}_e$, $\mathbf{b}_d$ : Bias vectors

$f$, $g$ : Activation functions (typically ReLU or sigmoid)

**Loss Function (Reconstruction Error):**

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N} \|x_i - \hat{x}_i\|^2$$

## 3. Clustering Techniques

1) **K-Means**: Applied to the encoded features to partition customers into distinct, non-overlapping groups based on purchasing patterns. The optimal number of clusters was determined using the elbow method and silhouette score

   **Objective Function:**

   $$\arg\min_{\{C_k\}} \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

   **Centroid Update:**

$$\mu_k^{(t+1)} = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$$

2) **Gaussian Mixture Model (GMM)**: Utilized for probabilistic clustering, capturing overlapping customer segments and complex cluster shapes in the latent space

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**Density Function:**

$$p(x) = \sum_{k=1}^{K} \alpha_k \mathcal{N}(x|\mu_k, \Sigma_k)$$

**E-step (Responsibility Calculation):**

$$\gamma(z_{nk}) = \frac{\alpha_k \mathcal{N}(x_n|\mu_k,\Sigma_k)}{\sum_{j=1}^{K} \alpha_j \mathcal{N}(x_n|\mu_j,\Sigma_j)}$$

**M-step (Parameter Update):**

$$\mu_k = \frac{\sum_n \gamma(z_{nk}) x_n}{N_k}$$

$$\Sigma_k = \frac{\sum_n \gamma(z_{nk})(x_n-\mu_k)(x_n-\mu_k)^T}{N_k}$$

$$N_k = \sum_n \gamma(z_{nk})$$

3) **DBSCAN**: Employed to identify dense regions and outliers, particularly effective for discovering irregular clusters and noise in compressed feature space

**ε-neighborhood:**

$$N_\varepsilon(p) = \{q \in D \mid \text{dist}(p, q) \leq \varepsilon\}$$

**Core Point Condition:**

$$|N_\varepsilon(p)| \geq \text{MinPts}$$

# 4. Performance Evaluation Metrics

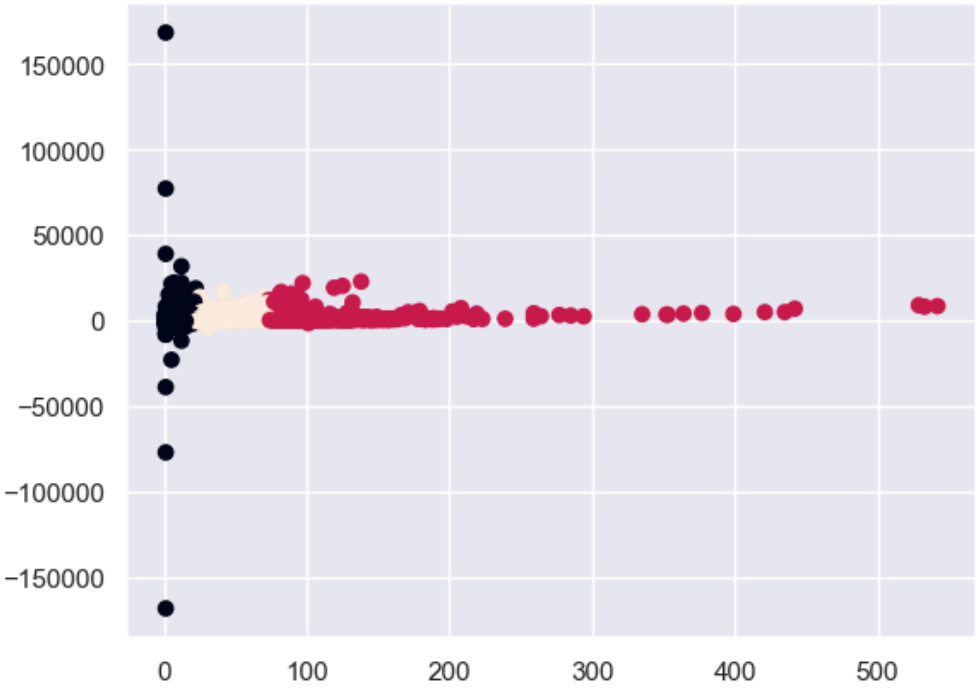
1) **Silhouette Score**: Evaluated the quality of clusters; higher scores indicate better-defined groupings.

2) **Cluster Visualization**: Used to qualitatively assess the separation and structure of customer segments in the reduced feature space.

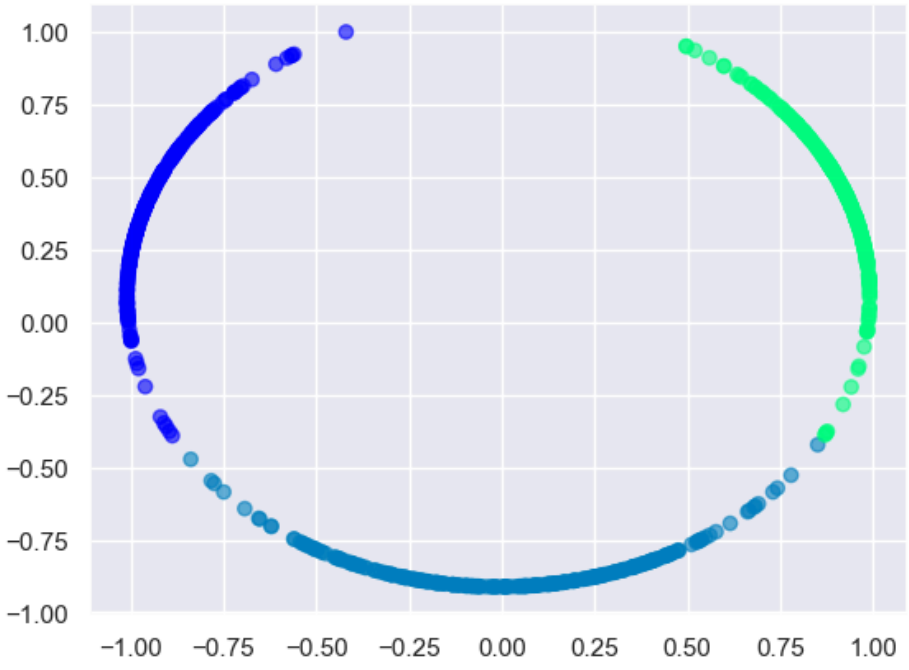
# Results

## 1. Before AutoEncoding-

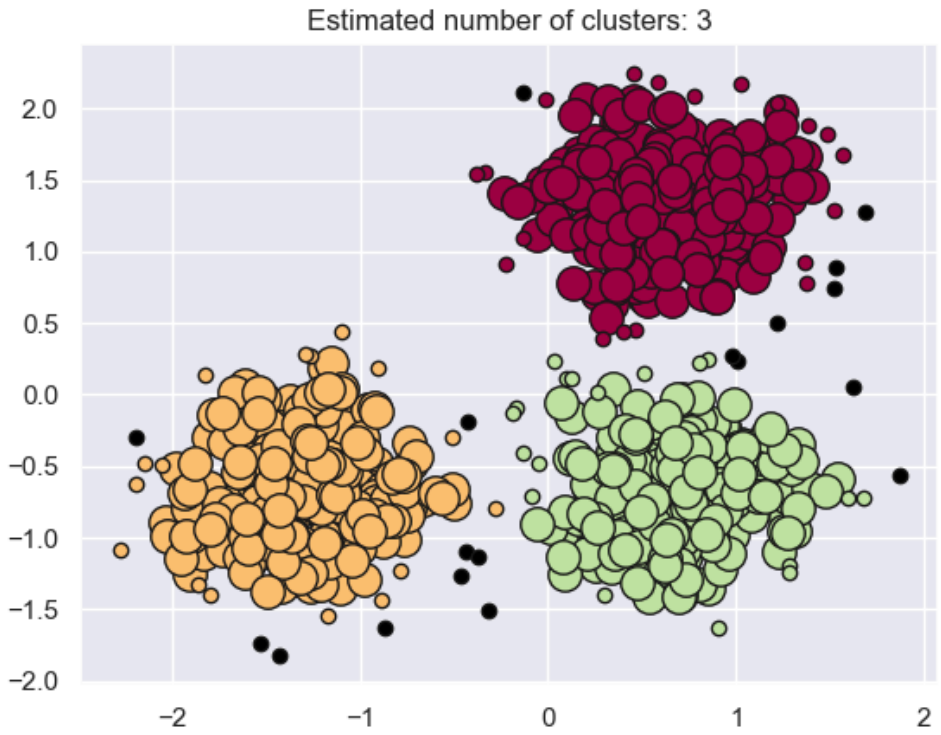
1) K-Means- Silhouette Score: 0.64




2) GMM- Silhouette Score: 0.76

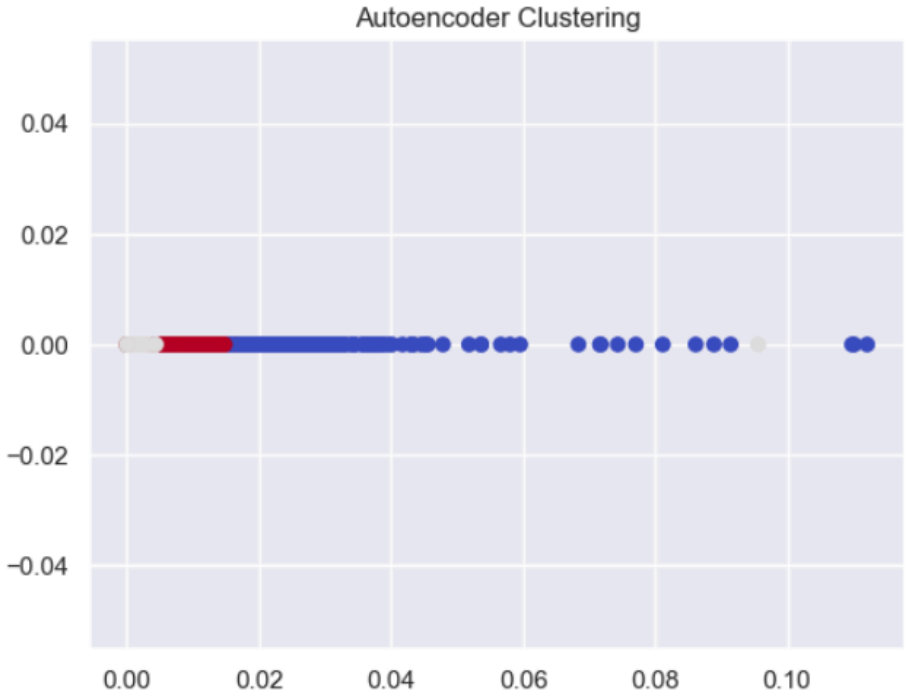


3) DBSCAN- Sillhoutte Score: 0.626

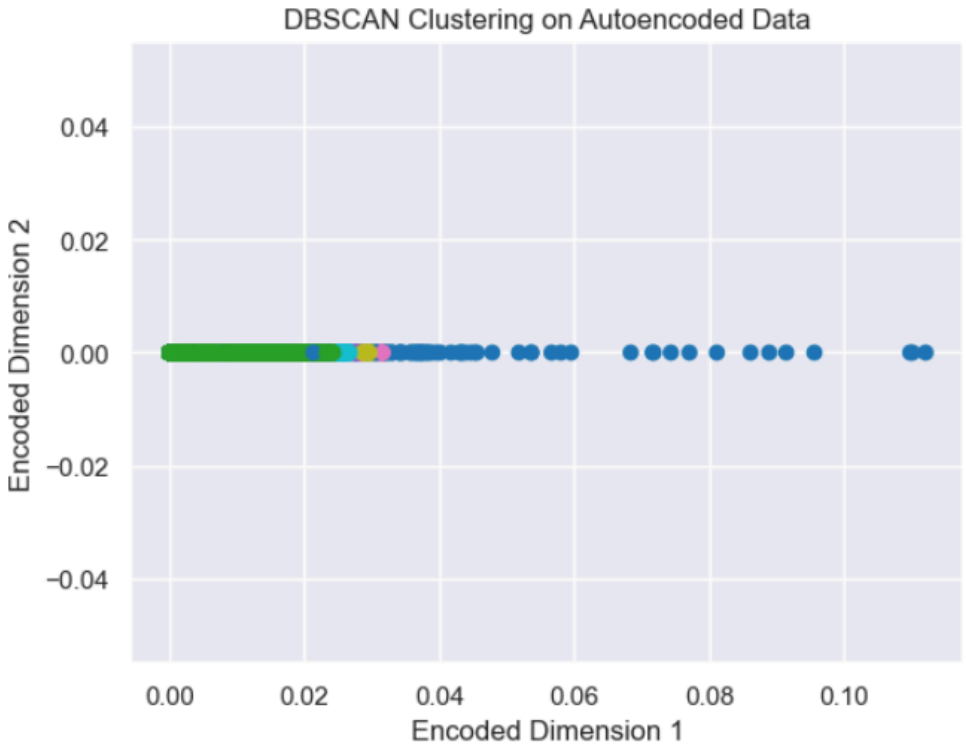



# 2. After Encoding-

1) K-Means- Silhouette Score:0.65



2) GMM- Sillhouette Score: 0.64
3) DBSCAN- Sillhouette Score: 0.79




## Summary:

| | Silhouette Score before AutoEncoding | Silhouette Score after AutoEncoding |
|---|---|---|
| K-Means | 0.64 | 0.65 |
| Gaussian Mixture Model | 0.76 | 0.64 |
| DBSCAN | 0.62 | 0.79 |

# References


1. John, Jeen & Shobayo, Olamilekan & Ogunleye, Bayode. (2023). An Exploration of Clustering Algorithms for Customer Segmentation in the UK Retail Market. Analytics. 2. 809-823. 10.3390/analytics2040042.

2. D. Chen. "Online Retail," UCI Machine Learning Repository, 2015. [Online]. Available: https://doi.org/10.24432/C5BW33.